

Flexible quadrilateral mesh of isogonal type in arbitrary size

YANG LIU* and JINSAN CHENG^{✉*}, Academy of Mathematics and System Science, CAS, China

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1 Examples and visualization

For $i = 1, 2, 3, 4$, the variation of $x_i := \tan(\frac{\alpha_i}{2})$ obey the so-called Bricard equation.

$$a_i x_i^2 x_{i+1}^2 + b_i x_i^2 + c_i x_{i+1}^2 + x_i x_{i+1} + e_i = 0, \quad (1)$$

where $x_5 := x_1$ and

$$\begin{cases} A_i := \cos(\mu'_i) - \cos(\lambda'_i + \gamma'_i + \delta'_i), \\ B_i := \cos(\mu'_i) - \cos(\lambda'_i + \gamma'_i - \delta'_i), \\ C_i := \cos(\mu'_i) - \cos(\lambda'_i - \gamma'_i + \delta'_i), \\ D_i := 4 \sin(\gamma'_i) \sin(\delta'_i) \neq 0, \\ E_i := \cos(\mu'_i) - \cos(\lambda'_i - \gamma'_i - \delta'_i), \\ (a_i, b_i, c_i, e_i) := \left(\frac{A_i}{D_i}, \frac{B_i}{D_i}, \frac{C_i}{D_i}, \frac{E_i}{D_i} \right). \end{cases}$$

We only consider antiisogonal meshes in which

$$\forall i, \lambda'_i + \mu'_i = \gamma'_i + \delta'_i = \pi,$$

so that $a_i = e_i = 0$ and Eq. (1) becomes

$$(kx_i - x_{i+1})(k'x_i - x_{i+1}) = 0$$

where $\{k, k'\} = \left\{ \frac{-1 \pm \sqrt{1-4b_i c_i}}{2c_i} \right\} = \left\{ \frac{\sin(\gamma'_i) \pm \sin(\lambda'_i)}{\sin(\gamma'_i - \lambda'_i)} \right\}$.

THEOREM 1. A 3×3 antiisogonal mesh in Fig. 1 is flexible if and only if

$$\forall i \in \{1, 2, 3, 4\}, \exists k_i \in \left\{ \frac{\sin(\gamma'_i) \pm \sin(\lambda'_i)}{\sin(\gamma'_i - \lambda'_i)} \right\} \text{ st. } \prod_{i=1}^4 k_i = 1.$$

Example 2 (General case). Set $k_i = \frac{\sin(\gamma'_i) - \sin(\lambda'_i)}{\sin(\gamma'_i - \lambda'_i)}$ and $K = k_1 k_2 k_3$, the 3×3 antiisogonal mesh in Fig. 1 is flexible if

$$\gamma'_4 = \arctan \left(\frac{(1 - K^2) \sin(\lambda'_4)}{(1 + K^2) \cos(\lambda'_4) - 2K} \right).$$

NOTE: The range of $\arctan$ is taken as $(0, \pi)$ and $\arctan(\infty) = \frac{\pi}{2}$.

Authors' Contact Information: Yang Liu, yangrio@amss.ac.cn; Jinsan Cheng, jcheng@amss.ac.cn, Academy of Mathematics and System Science, CAS, Beijing, China.

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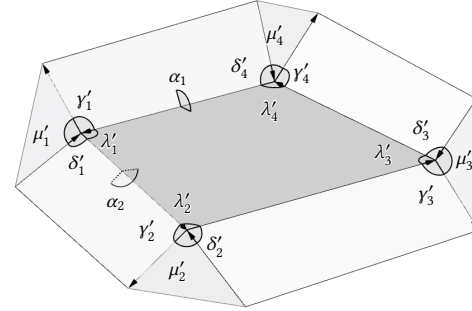


Fig. 1. Left: Sketch of a (trimmed) 3 by 3 p-mesh with fixed interior angles $\lambda'_i, \gamma'_i, \mu'_i, \delta'_i \in (0, \pi)$ of planar faces. Flexible angles α_i are dihedral angles between the central face and the attached faces. For i odd (even), α_i is measured above (beneath) the central face.

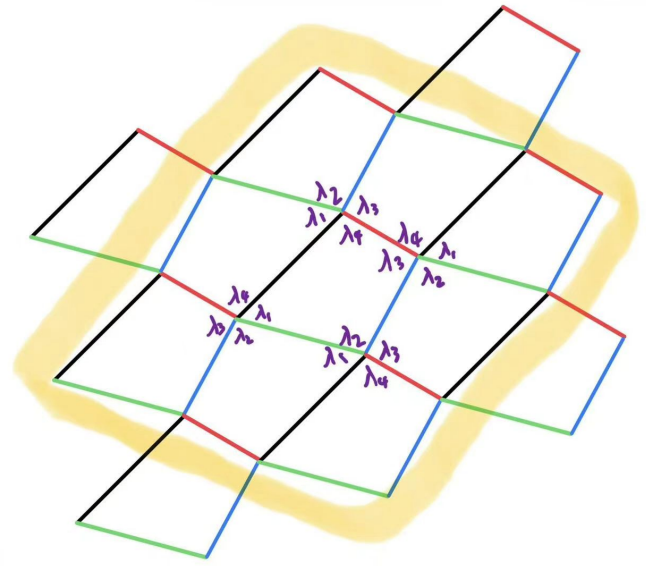


Fig. 2. All faces are congruent.

Example 3 (Congruent case). Choose $\lambda'_1, \lambda'_2 \in (0, \pi)$ but $\lambda'_1 \neq \lambda'_2$, and set

$$\lambda'_3 = \pi - \lambda'_1, \lambda'_4 = \pi - \lambda'_2, \gamma'_1 = \lambda'_4, \gamma'_2 = \lambda'_1, \gamma'_3 = \lambda'_2, \gamma'_4 = \lambda'_3.$$

Moreover, for $i \in \{1, 2, 3, 4\}$, set

$$k_i = \frac{\sin(\gamma'_i) - \sin(\lambda'_i)}{\sin(\gamma'_i - \lambda'_i)}, \quad K_i = \frac{\sin(\gamma'_i) + \sin(\lambda'_i)}{\sin(\gamma'_i - \lambda'_i)}.$$

One can check that $k_1 k_2 k_3 k_4 = K_1 K_2 K_3 K_4 = 1$.

References

A Research Methods

A.1 Part One

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A.2 Part Two

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B Online Resources

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